

$$\text{Rule 1: } \log_b (M \cdot N) = \log_b M + \log_b N$$

$$\text{Rule 2: } \log_b \left(\frac{M}{N} \right) = \log_b M - \log_b N$$

$$\text{Rule 3: } \log_b (M^k) = k \cdot \log_b M$$

$$\text{Rule 4: } \log_b (1) = 0$$

$$\text{Rule 5: } \log_b (b) = 1$$

$$\text{Rule 6: } \log_b (b^k) = k$$

$$\text{Rule 7: } b^{\log_b (k)} = k$$

Where:

$b > 0$ but $b \neq 1$, and M , N , and k are real numbers but M and N must be positive!

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